

Database Project Report Template

Due Date: 29/08/2026 - Week 3

Project ID & Title: #2: On-Demand Streaming Platform (Mini-Netflix)

A. Project Identity

- **Team Name:** No Man's Land (NML)
- **Team Members:** Nhan Duy Tran (n25dece106@student.ptithcm.edu.vn), Lam Vo (n25dece055@student.ptithcm.edu.vn), Hung Minh To (n25dece092@student.ptithcm.edu.vn)
- **Project Title:** Mini-Netflix On-Demand Streaming Platform: Managing Complex Content Hierarchies and Tiered Membership Access through EER Modelling and Relational Normalization

B. Report structure

1. *Introduction & Project Scope (Adapted from ISO/IEC/IEEE 29148)*

1.1 Business & Product Overview

1.1.1. Product Definition

The proposed system is an On-Demand Streaming Platform (Mini-Netflix). The platform is a centralized multimedia delivery service designed to manage, categorize, and stream digital video content to end-users.

To accurately manage this catalog, the system categorizes "Content" into two distinct hierarchical types:

Movies: Standalone video files characterized by a single duration and video resolution.

TV Series Episodes: Serialized media units that belong to a broader TV Series container, requiring complex tracking of seasons and episode numbers.

1.1.2. Customer & User Definition

The platform differentiates between administrative staff and end-user viewing through a strict hierarchical structure:

Database Administrator (DBA): The chief administrator who oversees and manages the database and DBMS resources. The DBA is strictly a backend role responsible for authorizing access, monitoring database use, coordinating hardware/software acquisition, and securing overall system performance.

Account Owner (The Customer): The master billing account responsible for administrative duties and purchasing the service. This entity acts as the primary "Customer" who manages the subscription tier (Basic, Standard, or Premium), handles billing status, and has the authority to create or delete sub-profiles.

Profile (The Viewer): The actual end-user entity partitioned under a parent Account. Viewing sessions cannot happen at the Account level; they must be done through a Profile. Profiles are restricted to viewing content and managing personal preferences (e.g., language, avatars). Profiles can be designated as a "Kids Profile," which restricts their access to specific age-appropriate rating classifications.

1.1.3. Value Proposition

The core business objective is to deliver a seamless and engaging user experience through three primary value pillars:

Personalized Viewing Experiences: By isolating viewing metrics into distinct M:N associative entities (Watchlist, Watch History, and Profile Ratings), the platform ensures that mutations or actions on Profile A never affect the recommendations or history of Profile B.

Tiered Access Control: The platform incentivizes upgrades by enforcing role-based content access. A user's subscription tier dictates their maximum video quality, the number of allowed concurrent streams, and access to premium catalog items.

Structured Content Discovery: Utilizing a classification lookup catalog for genres and strict metadata tracking (release years, ratings), the platform allows users to efficiently navigate large libraries of both standalone movies and serialized TV shows.

1.2. System Objective

1.2.1. Scenario Overview & Technical Environment

The proposed database system models the core backend architecture of an On-Demand Streaming Platform (Mini-Netflix). In this Universe of Discourse (UoD), the system orchestrates subscriber billing, multi-user profile separation, tiered content catalogs, active streaming sessions, and personalized watch metrics.

1.1.2 Core Purpose & System Objectives

The primary objective is to implement a robust, scalable, and normalized relational database schema (up to BCNF) that achieves:

- **EER Content Specialization:** Structuring a Content superclass with specialized Movie and TV Episode subclasses, avoiding null-heavy schemas.
- **Subscription Tier Access Control:** Enforcing role-based content access and concurrent stream limits directly via relational constraints.
- **Partitioned Profile Telemetry:** Isolating viewing histories, ratings, and watchlists per profile using resolved M:N associative entities.
- **Strict Referential Integrity:** Enforcing composite keys, foreign keys, and ON DELETE CASCADE policies to eliminate orphaned records.

1.3. Business Rules & Constraints

1.3.1 Data Elements & Structural Relationships

1.3.1.1 Subscription Tiers & Membership Plans

- **Subscription_Tier:** Defines membership levels, service quotas, and feature entitlements.

- Attributes: tier_id (PK, TINYINT, Check: 0, 1, 2, 3), tier_name (Unique: 'Free', 'Basic', 'Standard', 'Premium'), monthly_price, max_concurrent_streams, max_video_quality.

1.3.1.2 User Accounts & Profiles (1:N Structural Hierarchy)

- Account: Master administrative, billing, and credential-holding entity.
 - Attributes: account_id (PK), email (Unique), billing_status, current_tier_id (FK referencing Subscription_Tier.tier_id), pending_tier_id (FK referencing Subscription_Tier.tier_id, Nullable), pending_tier_effective_date (Nullable), registration_date, next_billing_date, canceled_at (Nullable), password_hash
- Profile: Viewing partition sub-entity isolated under a parent Account.
 - Attributes: profile_id (PK), account_id (FK referencing Account.account_id), profile_name, is_kids_profile (Boolean).
 - Cardinality: 1 Account has 1..5 Profiles (1:N, where $1 \leq N \leq 5$). Each Profile belongs to exactly 1 Account.
 - Semantic Invariant: UNIQUE(account_id, profile_name).

1.3.1.3 Content Catalog, EER Hierarchy & Media Units

- Catalog_Title (Superclass): High-level metadata container for all discoverable and browseable catalog assets.
 - Attributes: title_id (PK), title_name, maturity_rating (Check: 'G', 'PG', 'PG-13', 'R', 'NC-17'), minimum_tier_id (FK referencing Subscription_Tier.tier_id), content_type (Discriminator: 'Movie', 'Series'), synopsis
- Movie_Title (Subclass): Standalone feature films inheriting base catalog attributes.
 - Attributes: title_id (PK, FK referencing Catalog_Title.title_id), media_unit_id (FK referencing Streamable_Media_Unit.media_unit_id, Unique).
 - Hierarchy Constraint: {disjoint, complete} specialization from Catalog_Title.
- TV_Series (Subclass): Serialized multi-episode programs inheriting base catalog attributes.
 - Attributes: title_id (PK, FK referencing Catalog_Title.title_id), total_seasons.
 - Hierarchy Constraint: {disjoint, complete} specialization from Catalog_Title.
- Season: Intermediate structural grouping container for serialized TV series.
 - Attributes: season_id (PK), series_id (FK referencing TV_Series.title_id), season_number, release_year.
 - Cardinality: 1 TV_Series contains 1..* Seasons; each Season belongs to exactly 1 TV_Series.
 - Semantic Invariant: UNIQUE(series_id, season_number).
- Episode: Individual episodic unit belonging to a specific season.
 - Attributes: episode_id (PK), season_id (FK referencing Season.season_id), media_unit_id (FK referencing Streamable_Media_Unit.media_unit_id, Unique), episode_number, episode_title
 - Cardinality: 1 Season contains 1..* Episodes; each Episode belongs to exactly 1 Season.
- Streamable_Media_Unit: Concrete media file asset delivering actual video streams.
 - Attributes: media_unit_id (PK), duration_seconds, video_resolution, is_demo_or_trailer (Boolean), storage_asset_url.

1.3.1.4 User Activity, Telemetry & Session Entities (Resolved M:N)

- Watchlist (Associative M:N): Bookmarked media titles curated per profile.
 - Attributes: profile_id (PK, FK referencing Profile.profile_id), title_id (PK, FK referencing Catalog_Title.title_id), added_at (Timestamp).
 - Primary Key: (profile_id, title_id).
- Title_Rating (Associative M:N): Explicit user feedback and numerical review scores per title.
 - Attributes: profile_id (PK, FK referencing Profile.profile_id), title_id (PK, FK referencing Catalog_Title.title_id), score (TINYINT, Check: 1..10), rated_at (Timestamp).
 - Primary Key: (profile_id, title_id).
- Watch_History (Associative M:N Telemetry Log): Chronological viewing history and resumable playback checkpoints.
 - Attributes: profile_id (PK, FK referencing Profile.profile_id), media_unit_id (PK, FK referencing Streamable_Media_Unit.media_unit_id), watched_at (PK, Timestamp), playback_stop_seconds, is_completed (Boolean).
 - Primary Key: (profile_id, media_unit_id, watched_at).
- Active_Session: Real-time concurrent stream telemetry enforcing subscription device limits.
 - Attributes: session_id (PK, UUID), account_id (FK referencing Account.account_id), prof_id (FK referencing Profile.profile_id), ip_address, device_fingerprint, started_at (Timestamp), last_heartbeat_at (Timestamp), session_status.
 - Referential Integrity: Composite Foreign Key (account_id, prof_id) references Profile(account_id, profile_id) to maintain strict tenant scoping.
- **1.3.1.5 Administrative Security Admin_Audit_Log:** Secure backend trail for DBA mutations.
 - **Attributes:** log_id (PK), admin_user, action_type, table_affected, timestamp.

1.3.2. Operational Constraints & Semantic Integrity

1.3.2.1 Tier & Concurrency Rules

- BR-01 (Access Authorization): A Profile can only access Content where Content.required_tier_id <= Account.tier_id. (
- BR-02 (Stream Caps): Active concurrent records in Active_Session for a given account_id cannot exceed Subscription_Tier.max_concurrent_streams.

1.3.2.2 Profile Isolation & Parental Control

- BR-03 (Data Isolation): Watch_History, ratings, and Watchlist rows are strictly scoped to profile_id. Mutations on Profile A must never affect Profile B.
- BR-04 (Kids Filter): If Profile.is_kids_profile = TRUE, queries must exclude Content with rating_classification IN ('R', 'TV-MA').

1.3.2.3 Cardinality & Referential Bounds

- BR-05 (Profile Threshold): Count of Profiles per account_id must satisfy 1 <= Count <= 5.
- BR-06 (Cascade Protocol): ON DELETE CASCADE applies from Account -> Profile -> (Watchlist, Watch_History, Active_Session).

1.3.2.4 User Interaction & Service Access

BR-07 (Service Prerequisite): An unregistered user cannot interact with the streaming catalog. The creation of a valid Account is required to interact with the database.

BR-08 (Profile Enforcement): An Account cannot directly initiate a streaming session or save content. All viewing telemetry (Active_Session), Watch_History, and Watchlist records must be strictly tied to a Profile entity.

BR-09 (Demo/Free Tier Limitation): If an Account is registered but has an inactive billing_status or belongs to a "Free/Guest" tier, its associated Profiles are restricted to accessing Content categorized exclusively as 'Demo' or 'Trailer'.

1.3.2.5 Backend Security & Administrative Control

BR-10 (DBA Authorization): Only users authenticated with the Database Administrator (DBA) role have the privileges to execute Data Definition Language (DDL) commands, alter the core Content catalog, grant/revoke access permissions, or perform backend server maintenance.

BR-11 (Audit Tracking): All system-level mutations (adding new Movies/Episodes, deleting Accounts, updating Subscription Tier prices) performed by the DBA must be logged in a secure backend audit trail.

1.3.2.6 Billing & Subscription Lifecycle Constraints

BR-12 (Recurring Billing Cycle): The system mandates a recurring 30-day billing cycle starting from the Account.registration_date. The billed amount must strictly map to the monthly_price of the Account's currently assigned Subscription_Tier

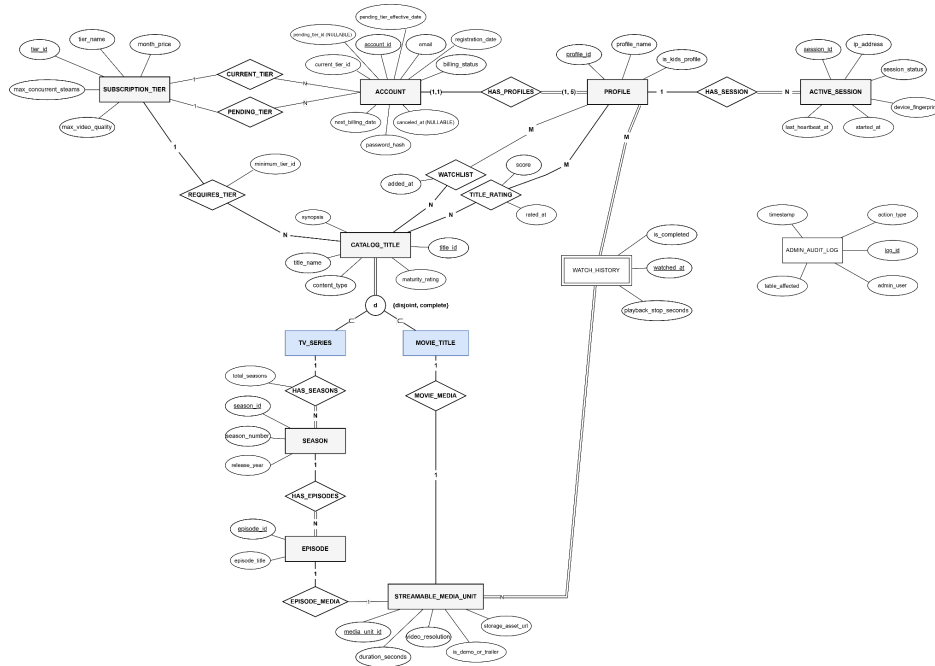
BR-13 (Delinquency & Suspension): If an Account fails to process a payment, its billing_status updates to 'Overdue' or 'Suspended'. When this occurs, the system must immediately revoke premium access for all associated Profiles, dropping their viewing privileges down to the "Demo/Free" constraints (as defined in BR-09).

BR-14 (Tier Modifications): If an Account Owner chooses to upgrade their Subscription_Tier (e.g., Basic to Premium), the change and new access limits take effect immediately. If the user downgrades, the change takes effect at the beginning of the *next* billing cycle.

BR-15 (Cancellation & Data Retention): Canceling a subscription sets the billing_status to 'Inactive'. The system does not immediately execute the ON DELETE CASCADE protocol. Instead, Profiles, Watchlist, and Watch_History records are retained for a 10-month grace period to allow the user to reactivate their account without losing their personalized data.

2. Database Design (ISO/IEC 19505 / IE Standards)

2.1 Conceptual Model (EER Diagram)



2.2 Logical Schema Mapping

2.2.1 Logical Relation Schema

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- Subscription_Tier: Defines membership levels, service quotas, and feature entitlements.
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- ### 2.2.2 IE Crow's Foot Schema

